

Lakshay Agarwal

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PROFESSIONAL SUMMARY

Backend Engineer with 3+ years of experience building scalable distributed systems using Java and Spring Boot. Specialized in designing high-throughput REST APIs, asynchronous event-driven architectures, AI-ready backend platforms, and high-availability caching tiers for applications serving millions of users. Experienced in distributed system design, high-level system design (HLD), optimizing low-latency services under peak concurrency, and building scalable cloud-native deployment infrastructure.

PROFESSIONAL EXPERIENCE

Goldman Sachs, Bengaluru, India

Jun 2025 – Present

Software Development Engineer

- Directed the core backend system and designed high-throughput inter-service communication using gRPC, performing back-of-the-envelope capacity planning to support 10,000+ requests per second.
- Reduced p99 API response latency by 45ms during peak traffic by implementing targeted Redis caching strategies and optimizing request lifecycles.
- Engineered zero-downtime deployment workflows by building comprehensive GitLab CI/CD pipelines from scratch, accelerating the project release lifecycle.
- Integrated robust telemetry systems to capture, monitor, and alert on critical real-time system metrics, improving overall backend observability.
- Architected and deployed highly available Redis infrastructure with Sentinel-based failover, achieving 99.99% uptime for critical backend financial services.
- Owned the migration of application secrets to HashiCorp Vault, establishing centralized secret management and strict role-based access control (RBAC).

Info Edge (Naukri.com), Noida, India

Jan 2023 – Jun 2025

Senior Software Development Engineer

- Led a zero-downtime migration of 10K+ client reports from MySQL to MongoDB to support highly unstructured data at scale while ensuring strict data consistency.
- Migrated 70% of legacy monolithic APIs to a Java Spring Boot microservices architecture, successfully serving 1M+ daily active users with sub-100ms latency.
- Designed Kafka-based asynchronous workflows for event-driven backend processing, decoupling core services and significantly increasing overall system fault tolerance.
- Optimized complex Elasticsearch reporting pipelines, reducing processing time by 57% and lowering compute overhead.
- Streamlined backend release processes using Jenkins and Docker CI/CD pipelines to ensure reliable automated deployments.

PROJECTS

Quma | Full-Stack E-Commerce Platform

Aug 2025 – Nov 2025

GitHub | *Java, Spring Boot, React, MongoDB, Redis, Elasticsearch, Razorpay API*

- Built a full-stack e-commerce platform from scratch using React and Spring Boot, implementing scalable REST APIs, product search, authentication, cart management, and payment workflows.
- Developed robust Razorpay payment workflows with idempotent APIs, webhook retry handling, and automated payment reconciliation to prevent duplicate transactions in distributed systems.
- Incorporated Redis caching, Elasticsearch-based product search, and database indexing strategies, reducing end-to-end API latency by 40% and improving overall platform responsiveness.

TECHNICAL SKILLS

- Languages & Frameworks:** Java, Spring Boot, Vert.x, React, Typescript, HTML, CSS, JavaScript, Python
- Architecture:** REST APIs, Microservices, Distributed Systems, Event-Driven Architecture, High-Level System Design (HLD)
- Databases & Messaging:** MySQL, MongoDB, Elasticsearch, Redis, Kafka
- DevOps & Infrastructure:** GitLab CI/CD, Docker, Kubernetes, Jenkins, Linux, Redis Sentinel,